

Model Card — Ultron-Qwen3.5-2B-ToolCall

Summary

Base model	unsloth/Qwen3.5-2B
Fine-tuned model name	Ultron-Qwen3.5-2B-ToolCall
Task	Structured tool/function calling for an autonomous penetration-testing agent
Fine-tuning method	QLoRA (4-bit base + low-rank adapters), via Unsloth Studio
Adapter merge	Merged and exported
Quantized release format	GGUF, Q6_K
License (base model)	Apache 2.0
License (this checkpoint)	Apache 2.0 (inherits base model license; custom dataset terms apply separately — see Dataset Card)
Intended use	Authorized penetration-testing / red-team lab agents that need to select and call tools (recon, scanning, reporting) from natural-language instructions
Out-of-scope use	Unauthorized or unsupervised offensive security actions; use against systems without explicit written authorization; general-purpose chat/assistant use (not optimized for it)

Model Details

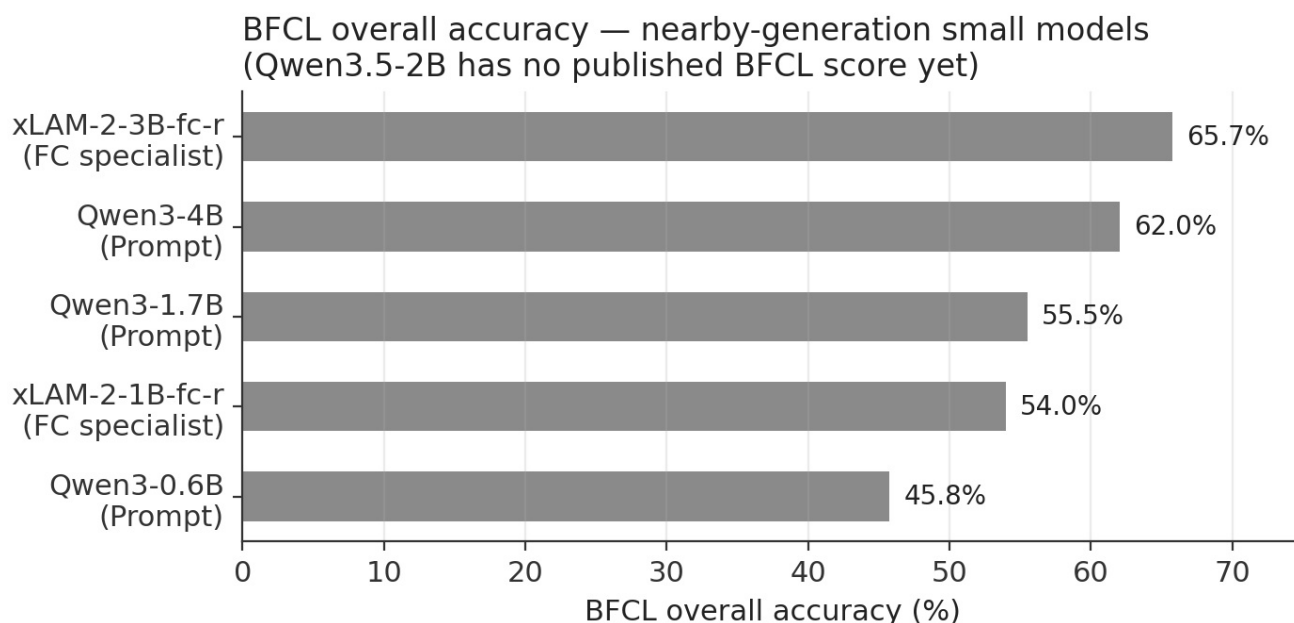
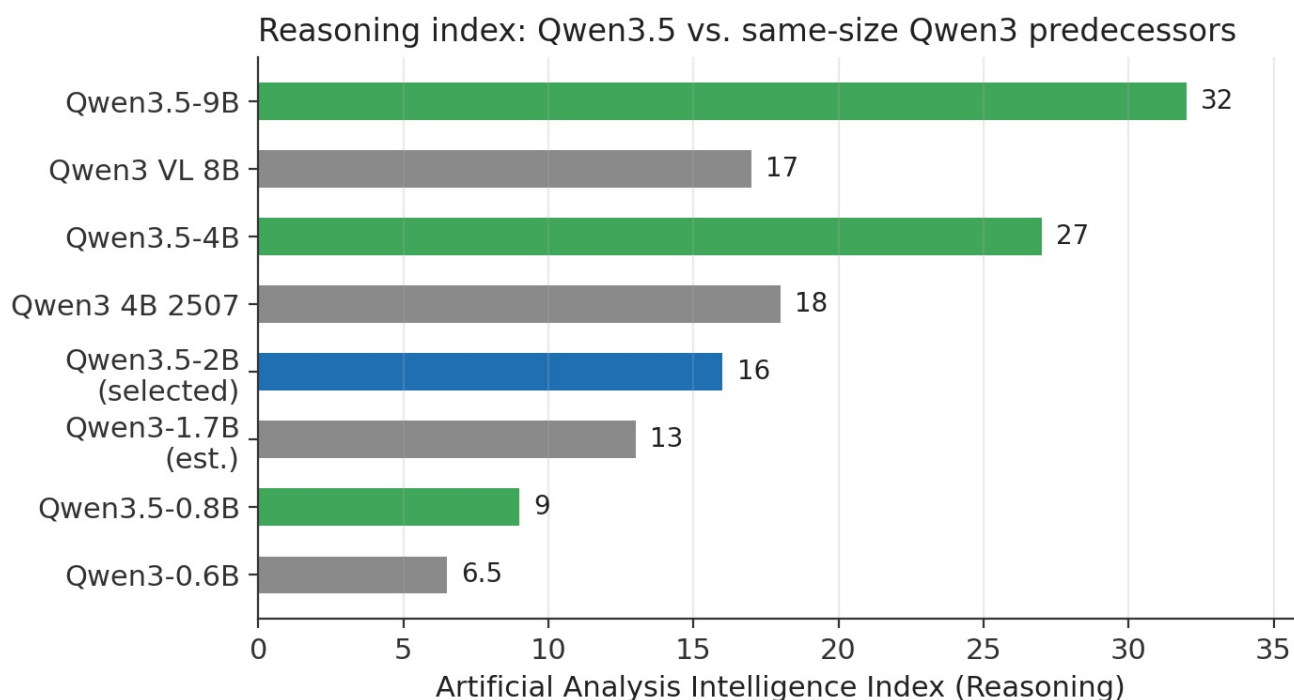
- **Architecture:** hybrid Gated DeltaNet + Attention, dense 2B-parameter transformer
- **Context length:** 262K tokens (native)
- **Training hardware:** single GPU, hackathon compute budget
- **Training time:** 2 minutes 14 seconds (30 steps, 40,580 tokens)
- **Final training loss:** 1.2061 (smoothed average 1.5623)
- **Final gradient norm:** 0.993
- **Peak learning rate:** $\approx 2 \times 10^{-4}$, linear warmup (steps 1-3) then linear decay to 8×10^{-6}

Full training curves and methodology are in `EVALUATION_REPORT.md` and the companion training report.

Why This Base Model

Qwen3.5-2B was selected over comparably sized alternatives (Qwen3-1.7B, Qwen2.5-1.5B, Llama-3.2-1B/3B, Gemma-3-4B-it, Phi-4-mini, SmolLM3-3B) for:

- Native OpenAI-compatible function-calling support
- Highest reasoning-per-parameter score at the 2B tier among tested peers (Artificial Analysis Intelligence Index)
- 262K native context, sufficient to hold a full tool manifest plus session/recon history
- Apache 2.0 license (unrestricted commercial/derivative use)
- Efficient hybrid architecture suited to continuous local inference alongside scanning tools



See the full comparison and benchmark charts in the project's model-selection report.

Training Data

Fine-tuned on a custom dataset of penetration-testing tool schemas and agent trajectories. See [DATASET_CARD.md](#) for composition, size, and provenance. **No public BFCL or other third-party benchmark data was used for training** — only the project's own custom dataset.

Evaluation Status

⚠ **No held-out evaluation set was configured for this training run.** Reported metrics (loss, gradient norm) describe fit to training data only. See [EVALUATION_REPORT.md](#) for the evaluation plan and current gaps before this checkpoint is used outside a supervised demo/lab setting.

Limitations

- Trained on a small dataset (40,580 tokens across a single 30-step run); has not been validated on held-out or adversarial

tool-calling prompts.

- Inherits any general-purpose limitations of the Qwen3.5-2B base model (may hallucinate, may follow a malformed instruction if not constrained by the orchestration layer).
- Is a general-purpose model adapted for tool calling, not a purpose-built function-calling specialist (e.g., xLAM-2) — accuracy on out-of-distribution tool schemas is unverified.
- Not evaluated for jailbreak robustness or prompt-injection resistance in the pentesting-agent context; the orchestration layer, not the model alone, must enforce authorization checks (see `RESPONSIBLE_USE_POLICY.md`).

Responsible Use

This model is designed to operate strictly within an authorized, scoped penetration-testing workflow, with human review before any state-changing or destructive tool call. See `RESPONSIBLE_USE_POLICY.md` for required guardrails before deployment.

Citation / Attribution

Base model: Qwen3.5-2B (Qwen team, Apache 2.0). Fine-tuning performed with Unsloth Studio for the "Ultron" hackathon project, August 2026.